



AUTONOMOUS ORBITAL BIOSCIENCE INFRASTRUCTURE

We aim to build and operate autonomous CubeSat BioLabs that transform antimicrobial drug-response testing in space—scalable access to high-quality biological data.

AUTONOMOUS EXECUTION

CREW-FREE PLATFORM

SCALABLE ARCHITECTURE

REPEATABLE ORBITAL R&D

A Nurse-Led Bioscience Research Satellite Project in Low Earth Orbit for Advancing Healthcare

Enabling Practical Orbital Bioscience

Microgravity may influence microbial adaptation, drug response, and biological stress behaviors in ways that cannot be fully replicated on Earth.

This platform is designed to enable more accessible, lower-barrier orbital experimentation and deliver mission-relevant biological data.

POTENTIAL APPLICATIONS

- Antimicrobial response testing
- Microbial risk monitoring
- Pharmaceutical pilot studies
- Translational biologic research
- Contract orbital R&D services

POTENTIAL USERS

- GOVERNMENT PROGRAMS
- RESEARCH UNIVERSITIES
- BIOTECH COMPANIES
- PHARMACEUTICAL PROGRAMS
- COMMERCIAL R&D INDUSTRY

Orbital Bioscience Needs a Scalable Operating Model

Space-based life science research remains operationally constrained by crew availability, limited flight cadence, and mission-specific integration complexity.

As demand grows for translational orbital bioscience, researchers and institutions need infrastructure that delivers meaningful biological data without requiring custom spacecraft programs or continuous crew interaction.

Autonomous Orbital Bioscience Infrastructure is being developed to address that gap through a scalable, crew-free wet-lab platform designed for repeatable orbital experimentation.

SCALABLE FIRST-MISSION ARCHITECTURE

Designed for autonomous data return first, with payload sample return as a future option for high-value use cases.

Data download first. High-value biological data returned each orbit.

Sample return later. Optional, on-demand capability for targeted missions.



AUTONOMOUS EXECUTION **FIRST** **LOWER OPERATIONAL BURDEN** **SAMPLE RETURN FUTURE OPTION**

Nurse-Led Leadership at the Frontier of Aerospace Bioscience

Interdisciplinary expertise in aerospace physiology, human systems integration, simulation, nursing, and translational space health research.

- Aerospace Physiology Expertise**
Specialized knowledge in human adaptation to microgravity, spaceflight stressors, and biological risk mitigation.
- NASA Twins Study Contributor**
Contributing to humanity's longest-running space biology study to understand human physiology in space.
 - Aerospace Physiology Research Excellence Award, Case Western Reserve University School of Medicine (April 29, 2026)
- Co-Author, NASA L'SPACE Funded Proposal**
Completed NASA L'SPACE Proposal Writing and Evaluation Experience (Spring 2025).
- Data Analyst & Simulation Specialist**
Served as Data Analyst and Simulation Specialist for a NASA L'SPACE project.
- Aerospace Nurse Researcher (PI/Co-PI)**
Leading multiple spaceflight research proposals as PI/Co-PI, including a Philippine 3U biological CubeSat concept and projects on space radiation protection and life support systems using bryophytes and fungi species; previously led the 2025 IAS Flight Opportunities proposal on multivitamin absorption in suborbital flight.



Dr. J. P. Hernandez
Nurse Educator, Scientist, Futurist, & Entrepreneur

CORE STRENGTHS

SCIENCE-DRIVEN DECISION MAKING

CROSS-DISCIPLINARY COLLABORATION

TRANSLATIONAL RESEARCH FOCUS

MISSION-FIRST LEADERSHIP

INNOVATION FOR HUMANITY

INNOVATING TECHNOLOGIES AND COUNTERMEASURES FOR SPACE HEALTH. ADVANCING NURSING. ELEVATING HUMANITY.

Most Orbital Life Science Missions Still Depend on High-Touch Operations

Traditional workflows remain expensive, operationally intensive, and difficult to scale.

- CREW DEPENDENCE
- ISS SCHEDULING
- CUSTOM INTEGRATION
- HIGH OPERATIONAL OVERHEAD
- DIFFICULT REPEATABILITY



Microgravity can alter microbial behavior and drug response in ways that are difficult to reproduce terrestrially. Yet access to reliable orbital experimentation remains limited.



10⁻⁶ m/s²
MICROGRAVITY LEVEL



420 km
TYPICAL LEO ALTITUDE



51.6°
TYPICAL ORBIT INCLINATION



THE RESULT:
Scientifically valuable experiments are delayed, underscaled, or not attempted at all.



WE NEED A BETTER WAY
Autonomous. Scalable. Repeatable. Built for orbital life science.

From Pilot Missions to Repeatable Orbital R&D Infrastructure

The long-term vision extends beyond single experiments toward recurring mission services and modular orbital bioscience capability.



STANDARDIZED ASSAY PACKAGES

MODULAR WET-LAB SERVICES

HOSTED PAYLOAD INTEGRATION

RECURRING FLIGHT CAMPAIGNS

CONTRACT ORBITAL R&D SERVICES



MODULARITY



SCALABILITY



ACCESSIBILITY



PARTNERSHIP ECOSYSTEM

Interchangeable wet-lab and sensor modules enable rapid reconfiguration for diverse research objectives.

From single CubeSats to distributed constellations, building a scalable orbital lab network.

Lower cost, more frequent access to microgravity accelerates bioscience discovery and innovation.

Open architecture for academia, industry, and government collaboration on future health breakthroughs.

BUILT FOR MODULARITY. DESIGNED FOR SCALE.

REVOLUTIONIZING CUBESAT CAPABILITIES IN SPACE RESEARCH

04 THE SOLUTION

A Free-Flying Orbital Wet-Lab Platform

Our CubeSat BioLab integrates sealed microfluidics, timed dosing, onboard sensing, environmental monitoring, and data logging into a compact, self-operating platform.



BOLAB-X1
AUTONOMOUS BIOSCIENCE CUBE LAB

TIMED DOSING

DATA LOGGING & STORAGE

AUTONOMOUS EXECUTION

ONBOARD SENSORS

ENVIRONMENTAL MONITORING

CORE CAPABILITIES

MICROFLUIDIC ASSAY MODULE
Enables cell & microbial experiments in a sealed, contamination-controlled environment.

TIMED DOSING
Precision fluid handling with programmable intervals for dynamic experiment control.

ONBOARD SENSORS
Monitors temperature, pressure, radiation, humidity, and vibration in real time.

ENVIRONMENTAL MONITORING
Maintains optimal conditions and logs environmental data throughout the mission.

DATA LOGGING & STORAGE
Secure onboard storage with time-stamped data packets and health telemetry.

AUTONOMOUS EXECUTION
Self-operating workflows enable experiments without crew interaction and complexity.

KEY BENEFITS

RISK REDUCTION
Sealed, autonomous design minimizes contamination and operational risk.

COST EFFICIENCY
Lower mission cost through autonomy and reduced crew-time requirements.

HIGH DATA YIELD
Continuous experiments provide richer, time-resolved biological datasets.

MODULAR & SCALABLE
Standardized modules support rapid iteration and multi-mission deployments.

REAL-TIME INSIGHTS
Download enables fast decision-making and adaptive experiment control.

TRANSFORMATIONAL
Advances life science discovery and enables new frontiers in space health research.

ORBITAL BIOSCIENCE. ADVANCING LIFE SCIENCE IN SPACE.
Smarter Experiments. Lower Cost. Greater Impact.

09 INVESTMENT & PATH FORWARD

A Strategic Entry Point into Orbital Bioscience

We are advancing technical de-risking, payload architecture definition, assay validation planning, and commercialization preparation to enable future orbital demonstration.

CURRENT FOCUS AREAS

Microfluidic prototyping

Systems integration

Biosafety planning

Environmental testing

Assay validation

Mission architecture definition



BOLAB-X1
AUTONOMOUS BIOSCIENCE CUBE LAB

INVESTMENT NEEDED FOR CURRENT PROJECT PHASE

\$250K

Milestone-based technical de-risking phase

05 MISSION ARCHITECTURE

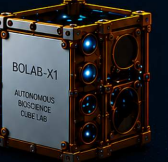
Software-Defined Architecture Changes the Economics of Orbital Bioscience

Our architecture reduces infrastructure dependence and operational burden while enabling repeatability, modularity, and future service scalability.

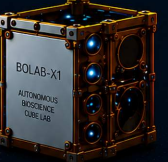
- EXPERIMENT DESIGN & PAYLOAD INTEGRATION
- LAUNCH & DEPLOYMENT
- CREW-FREE ORBITAL OPERATION
- DATA DOWNLINK & ANALYSIS
- SPACECRAFT DEORBIT END OF MISSION

HOW IT WORKS

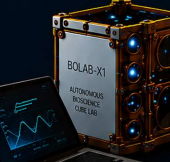
1. GROUND INTEGRATION
Experiments prepared. BoLab integrated & tested.



2. ORBITAL OPERATIONS
Self-operating execution. Real-time science.



3. MISSION OUTCOMES
Actionable data. Reviewed samples. Deeper insights.



10 LET'S BUILD THE FUTURE

Building the Infrastructure Layer for Orbital Bioscience

A modular, autonomous, and intelligent spacecraft infrastructure designed to make orbital bioscience missions accessible, repeatable, and scalable.



BOLAB-X1
AUTONOMOUS BIOSCIENCE CUBE LAB

Spacecraft bus
Attitude control, command & data handling, power distribution, thermal control.

Sealed biosafety cartridge
Aseptic, single-use cartridges for dosing & antimicrobial containment.

Optical sensing module
Onboard microscopy & detection for growth/inhibition measurement and microbial analysis.

Environmental monitoring
Temperature, humidity, pressure, and gas sensors.

Onboard data logging
Data capture, time-stamping, and secure non-volatile storage.

Endpoint preservation chamber
Stabilizes and preserves samples for return to Earth.

Communications & power subsystem
S-band communications, power conversion, and battery storage.

WHY IT MATTERS

- Reduces mission cost through modular reuse.
- Enables frequent, low-cost orbital bioscience missions.
- Standardized architecture accelerates time to orbit.
- Supports diverse experiments across institutions and industries.
- Paves the way for a commercial orbital bioscience ecosystem.

LET'S CONNECT
Dr. Joannes Paulus Tolentino Hernandez
Project Team Leader

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